

FusionForests

Data fusion for causal survival inference — project update

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What. A Bayesian nonparametric framework for fusing a randomised trial (RCT) with an observational study (OS) under unmeasured confounding in the OS. Implemented in the `FusionForests` R package.

Two explicit goals

- (G1) Efficiency gain** for heterogeneous treatment-effect estimation, by borrowing from the OS.
- (G2) Identifiable extrapolation** of causal survival contrasts beyond the RCT follow-up horizon, using the longer follow-up of the OS.

Niche. Existing methods address confounding correction *or* long-term extrapolation, rarely both, and almost none with a flexible nonparametric error. The aim is all three at once.

AFT decomposition ($Y_i = \log T_i$):

$$Y_i = m_0(X_i, S_i) + A_i \tau(X_i) + (1 - S_i) A_i c(X_i) + \sigma \varepsilon_i,$$

with m_0, τ, c each modelled by BART forests and $\varepsilon \sim P$ under a hierarchical-DP prior. τ is identified from the RCT; c is identified from the OS once τ is anchored.

Three causal contrasts on the survival scale

- $\Delta_{\text{SD}}(t, x)$ — survival-probability gain at a fixed horizon t .
- $\Delta_{\text{RMST}}(t^*, x)$ — expected life-time gain on $[0, t^*]$; the headline summary under (G2).
- $\text{AF}(x) = \exp\{\tau(x)\}$ — acceleration factor: horizon-free, HDP-independent, causally interpretable under frailty and effect heterogeneity (Brathovde et al. 2024).

Methodology

- **HDP nonparametric error** — five modes from Gaussian through to the full hierarchical-DP with per-source recentring. The HDP fixes a centring pathology that destabilised the pooled DP under bimodal OS noise.
- **Causal estimands** — closed forms for Δ_{SD} and Δ_{RMST} under the HDP; full argument for the acceleration factor as the principal causal target.
- **Posterior projection summaries** (Woody–Carvalho–Murray) — BART posterior of τ or the AF projected onto a small interpretable basis; uncertainty inherited exactly.
- **Framework writeup** — identification, transportability, structural decomposition of the confounding function.

Application

- ACTG175 (RCT) + MACS (OS): first end-to-end HIV survival analysis underway.

Prior milestone. Horseshoe Forests paper, arXiv:2507.22004.

Remaining challenges and next steps

Methodological

- *Constancy of τ over time*: waning or late-emerging effects break extrapolation.
- *RCT does double duty*: anchors τ and disambiguates τ from c — fragile when n_{RCT} is small.
- Calendar-time effects in long OS; informative OS censoring; covariate overlap at long t .

Implementation. HDP combined with per-source scale; adaptive truncation; DP-aware censored augmentation; cure-fraction; time-varying $\tau(x, t)$.

Next

- Finish the HIV case study; simulations over $n_{\text{RCT}}/n_{\text{OS}}$ and extrapolation horizons.
- Paper draft at *causal fusion* $\cap$ *long-term extrapolation* $\cap$ *nonparametric error*.